■ Splunk SIEM Lab

Security Monitoring & Log Analysis

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Splunk Enterprise 10.2.2 • Host: LAPTOP-HM3MTRGG • Port 9997

■ Lab Environment

SIEM Platform	Splunk Enterprise 10.2.2 for Windows
Host Machine	LAPTOP-HM3MTRGG
Web UI	localhost:8000
Receiving Port	9997 (enabled for log forwarding)
Indexes Used	main, hoemlab, linux, sample_auth_log_1
Log Sources	Windows Event Logs, Perfmon, tutorialdata.zip, auth.log, sample_auth_log.txt
Sourcetypes	WinEventLog:Security, Perfmon:CPU Load, Perfmon:Network Interface, Linux_auth, Linux Syslog, www1/secure

■ Objectives

- Install Splunk Enterprise 10.2.2 on Windows and access the Web UI at localhost:8000
- Configure receiving port 9997 for log forwarding
- Ingest multiple log sources: Windows Event Logs, Perfmon metrics, tutorialdata, Linux auth logs
- Write and execute SPL queries to search and filter security events
- Identify SSH brute-force and credential stuffing patterns in log data
- Investigate Windows Security Event Viewer and correlate with Splunk findings
- Understand SPL case sensitivity and its impact on detection queries

Step 1 — Splunk Installation & Web UI Access

Downloaded and installed **Splunk Enterprise 10.2.2 for Windows**. Launched Splunk and accessed the Web UI at **localhost:8000**. Logged in as Administrator. The home dashboard confirmed Splunk Enterprise was running with Search & Reporting, Audit Trail, Splunk Secure Gateway, and Upgrade Readiness App installed.

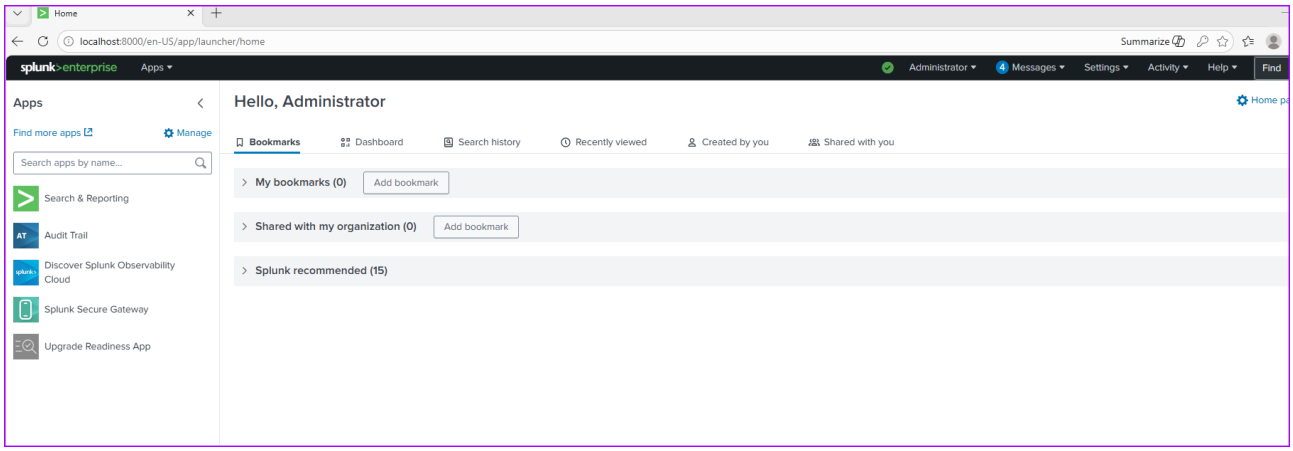

Figure 1 — Splunk Enterprise home: 'Hello, Administrator' dashboard with installed apps

Step 2 — Configuring Settings & Receiving Port

Navigated to the **Settings** menu to explore configuration options. Confirmed access to Data inputs, Forwarding and receiving, Indexes, Source types, and Users & Authentication.

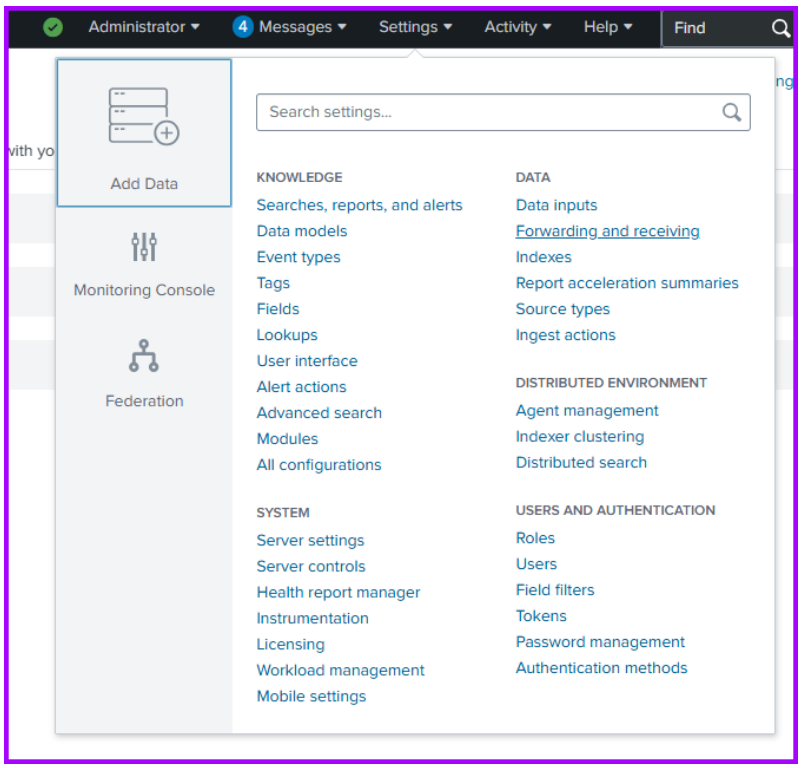

Figure 2 — Splunk Settings menu showing all configuration categories

Navigated to **Settings** → **Forwarding and Receiving** → **Receive data**. Created a new receiving port: **9997**. Confirmed the port was saved and showing status: **Enabled**.

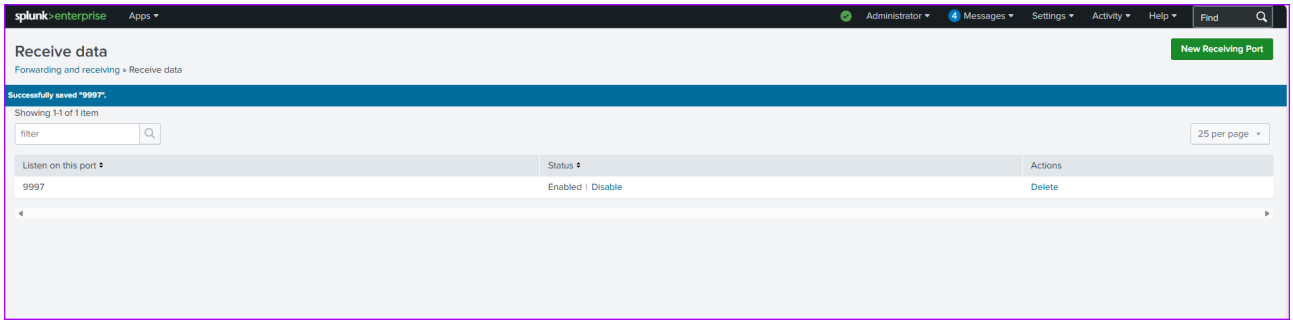

Figure 3 — Receive data page: port 9997 successfully saved and Enabled

Step 3 — Ingesting Data Sources

Data Source — tutorialdata.zip

Used **Settings** → **Add Data** to upload log data into Splunk. The Add Data wizard offered three ingestion methods: Upload, Monitor, and Forward.

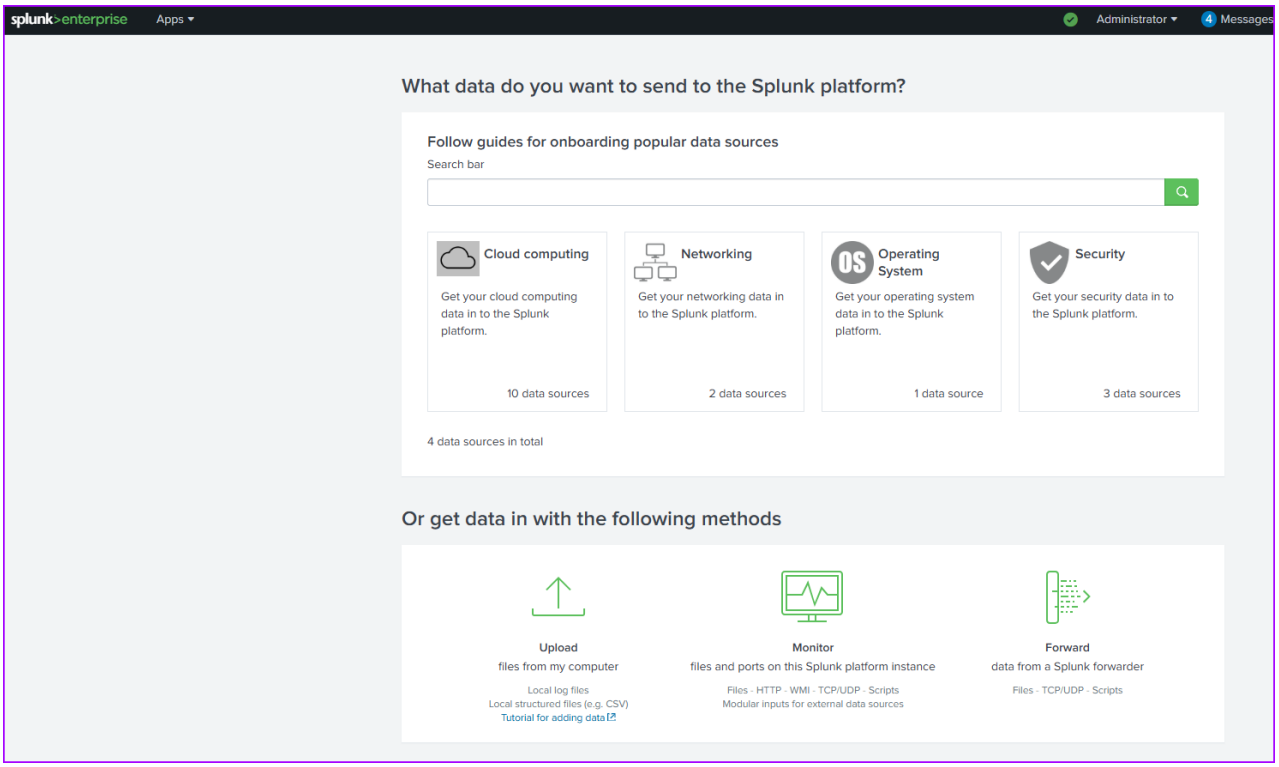

Figure 4 — Add Data wizard: Upload, Monitor, Forward options for data ingestion

Configured the input settings for the upload:

Source Type	Automatic
Host	LAPTOP-HM3MTRGG (Constant value)
Index	hoemlab

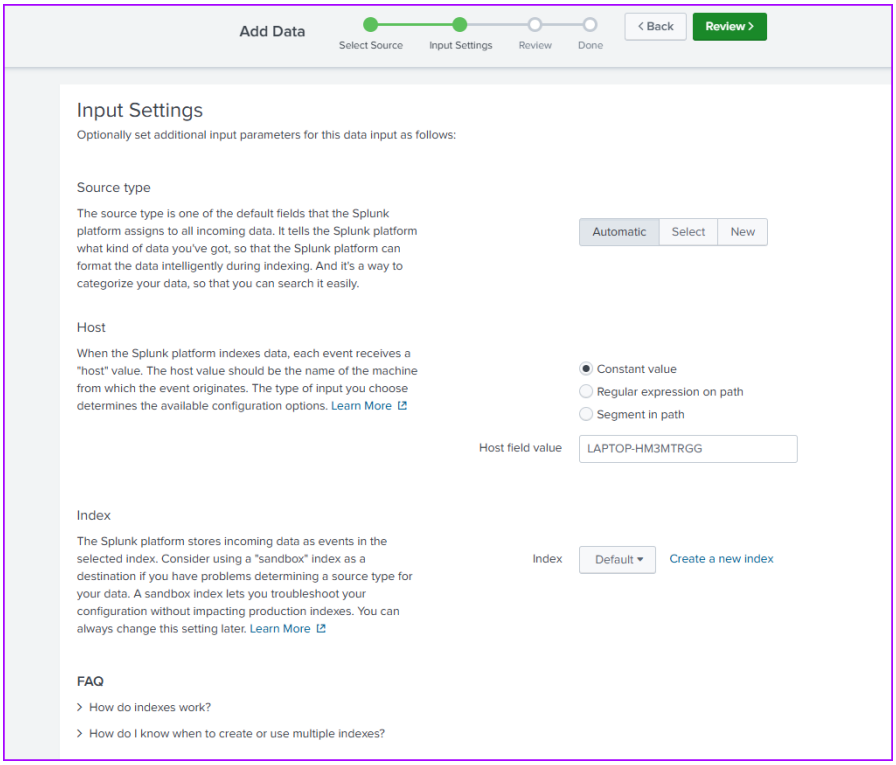

Figure 5 — Input Settings: Host field = LAPTOP-HM3MTRGG, sourcetype Automatic

Reviewed the upload configuration before submitting:

Input Type	Uploaded File
File Name	tutorialdata.zip
Source Type	Automatic
Host	LAPTOP-HM3MTRGG
Index	hoemlab

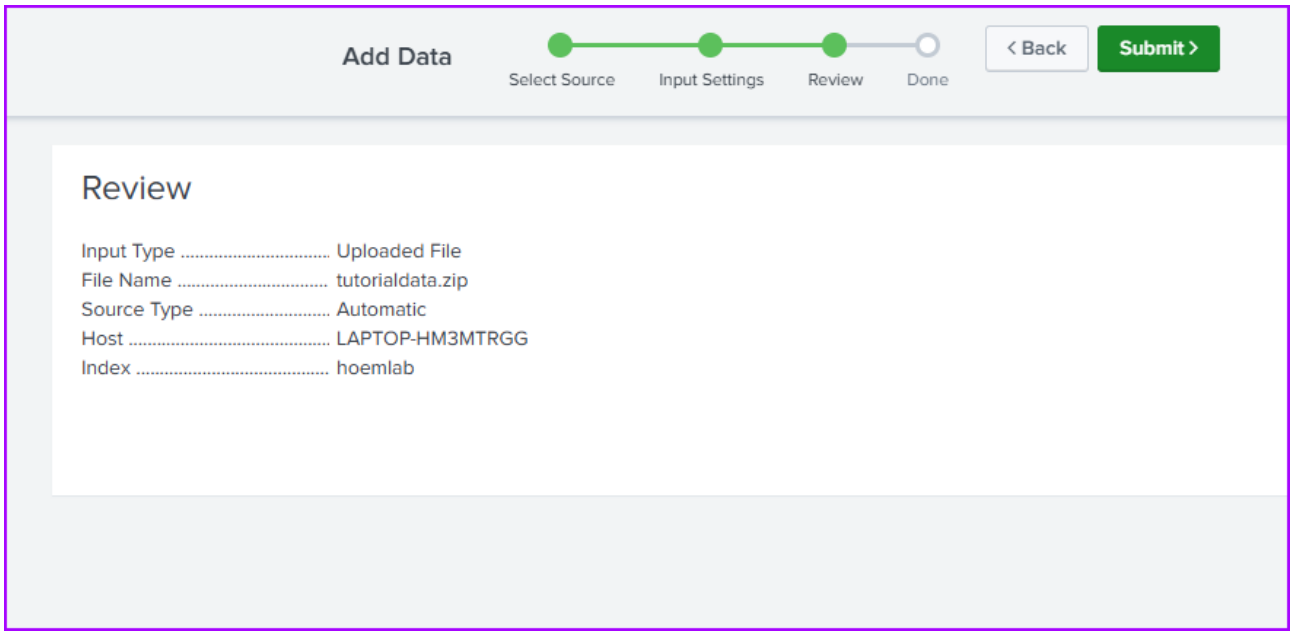

Figure 6 — Add Data Review page confirming tutorialdata.zip settings before submit

Submitted the upload. Splunk confirmed: **'File has been uploaded successfully.'** Clicked 'Start Searching' to begin querying the new data.

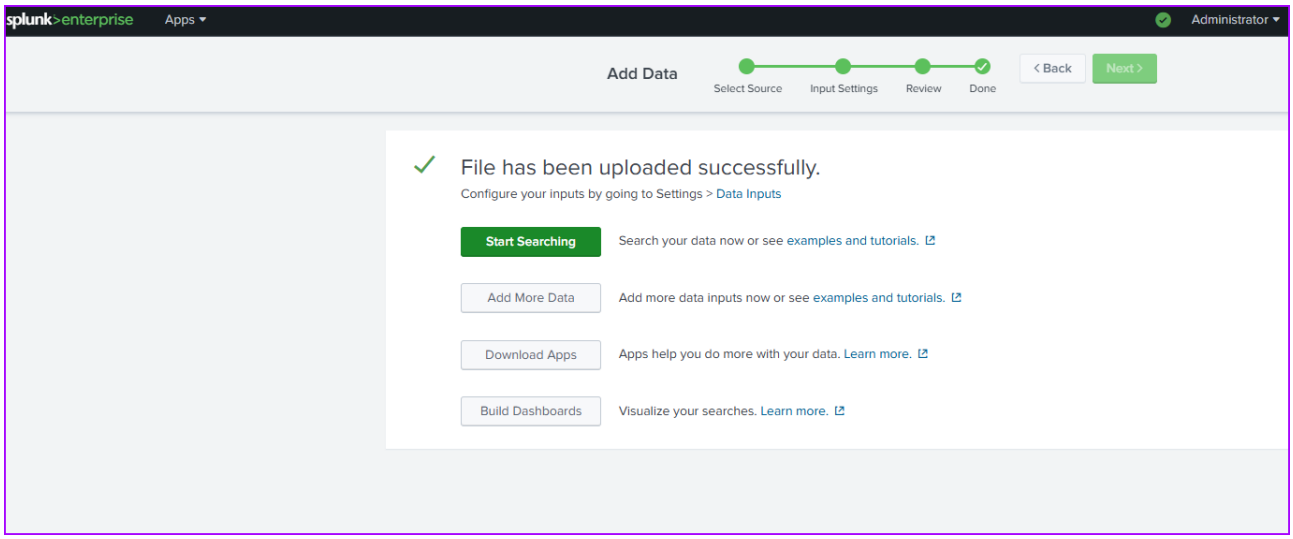

Figure 7 — 'File has been uploaded successfully' confirmation page

Step 4 — SPL Queries & Log Analysis

Search 1 — Verify host data (Windows Event Logs + Perfmon)

```
host="LAPTOP-HM3MTRGG"
```

Returned **669 events** (4/5/26 7:00 AM – 4/6/26 7:28 AM). Events included Perfmon: Network Interface (Bytes Sent/sec, Bytes Received/sec) and Perfmon: CPU Load (% User Time). Confirmed multiple sourcetypes ingesting correctly.

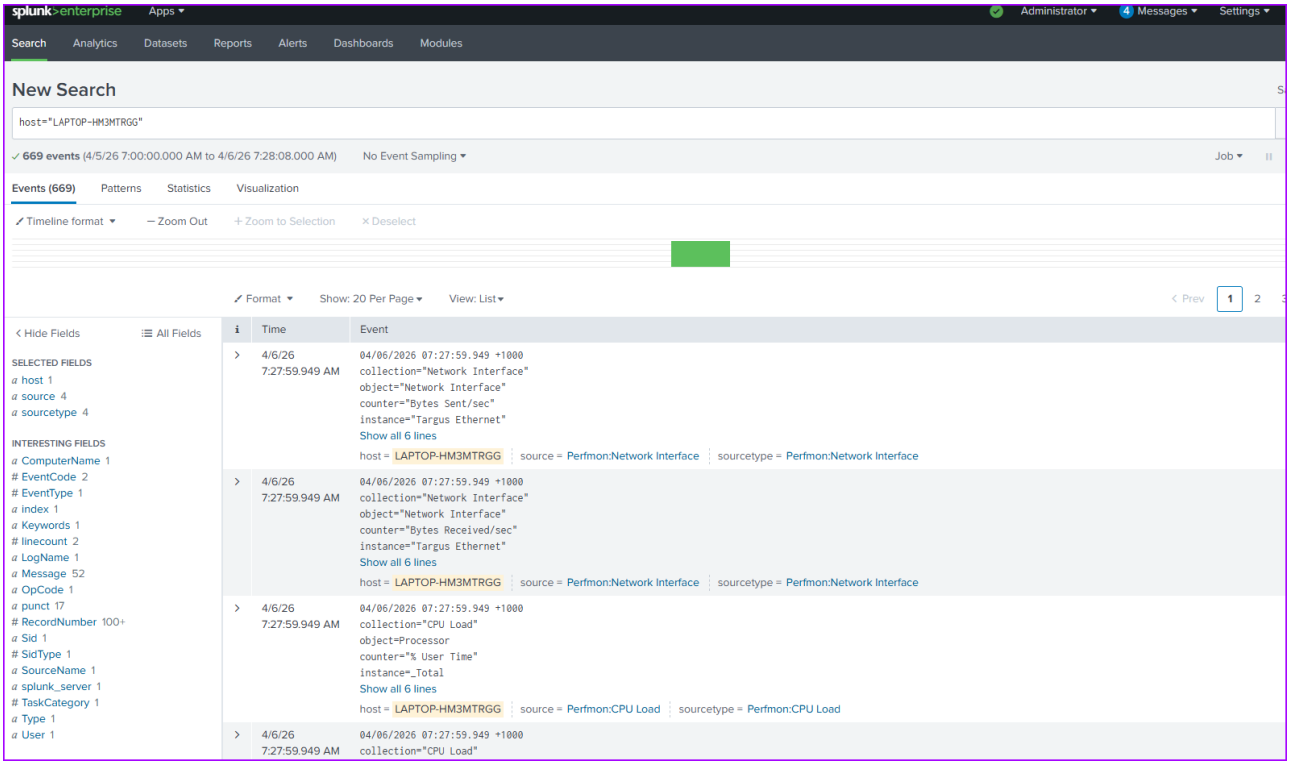

Figure 8 — 669 events for host=LAPTOP-HM3MTRGG showing Perfmon Network Interface and CPU Load data

Search 2 — All events in hoemlab index

```
source="tutorialdata.zip:*" host="LAPTOP-HM3MTRGG" index="hoemlab"
```

Returned **109,864 events**. Events showed repeated SSH failed password attempts via mailsv1 sshd for multiple invalid users (appserver, root, testuser, apache, mongodb, mail, games) all originating from IP **194.8.74.23**. Source: tutorialdata.zip:\mailsv\secure.log, sourcetype: www1/secure.

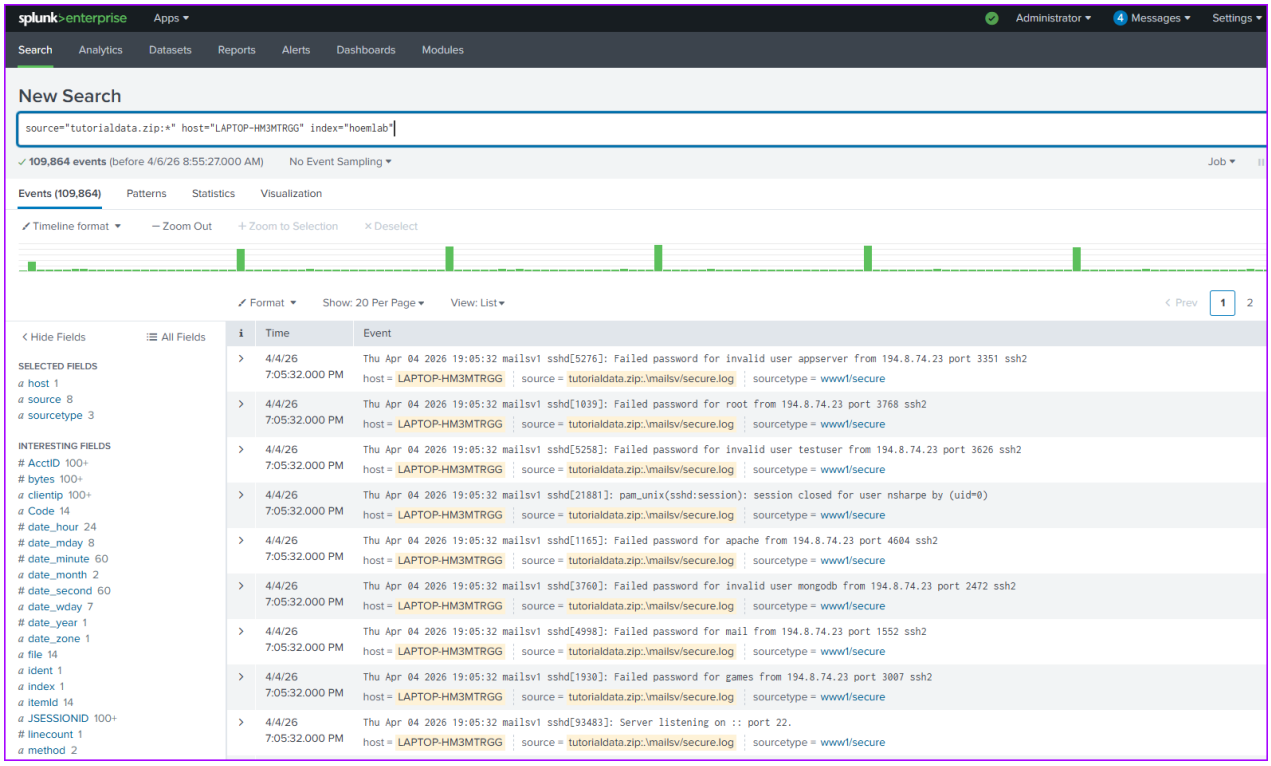

Figure 9 — 109,864 events in hoemlab index: SSH failed passwords from 194.8.74.23 targeting multiple accounts

Search 3 — Filter for failed events (brute-force pattern)

```
source="tutorialdata.zip:*" host="LAPTOP-HM3MTRGG" index="hoemlab" fail*
```

Filtered to **33,253 events** matching fail*. Results showed repeated SSH failed password attempts for multiple invalid users from the same attacker IP 194.8.74.23 — a classic brute-force pattern targeting service accounts.

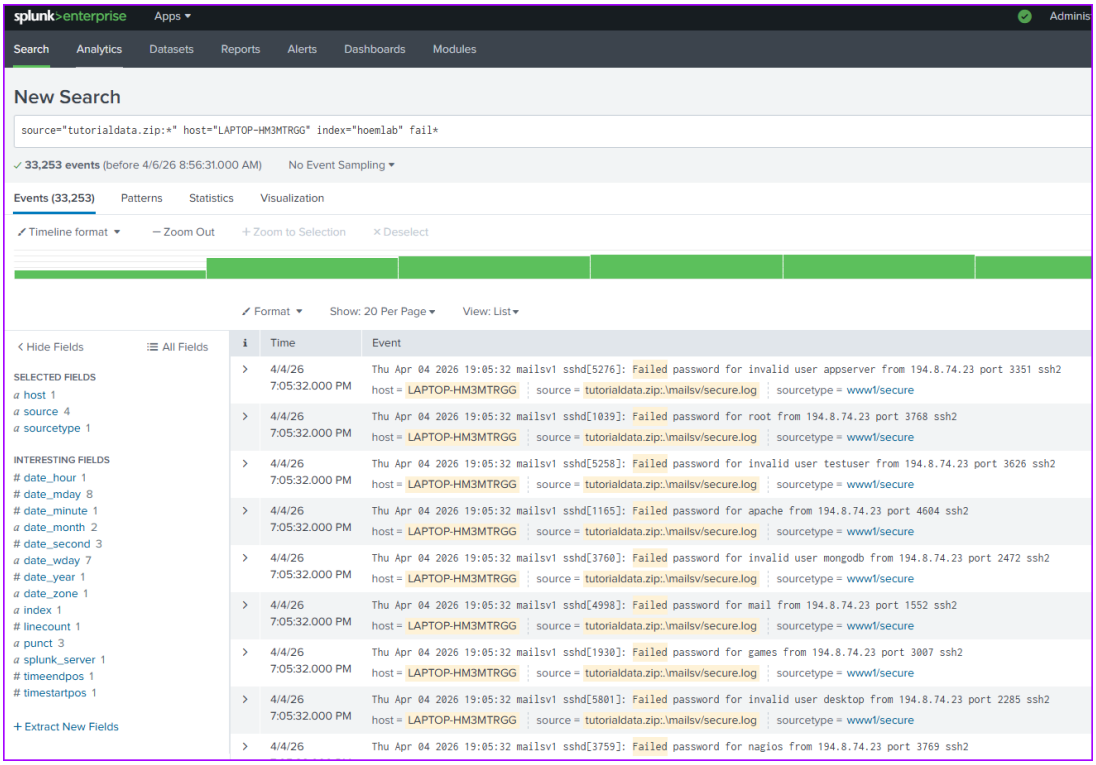

Figure 10 — 33,253 events matching fail showing SSH brute-force pattern from 194.8.74.23*

Search 4 — SSH-specific events

```
source="tutorialdata.zip:*" host="LAPTOP-HM3MTRGG" index="hoemlab" ssh*
```

Returned **38,950 events** matching ssh*. Mix of failed password attempts and legitimate session events (pam_unix session closed for nsharpe). Showed both attack traffic and normal session activity in the same log.

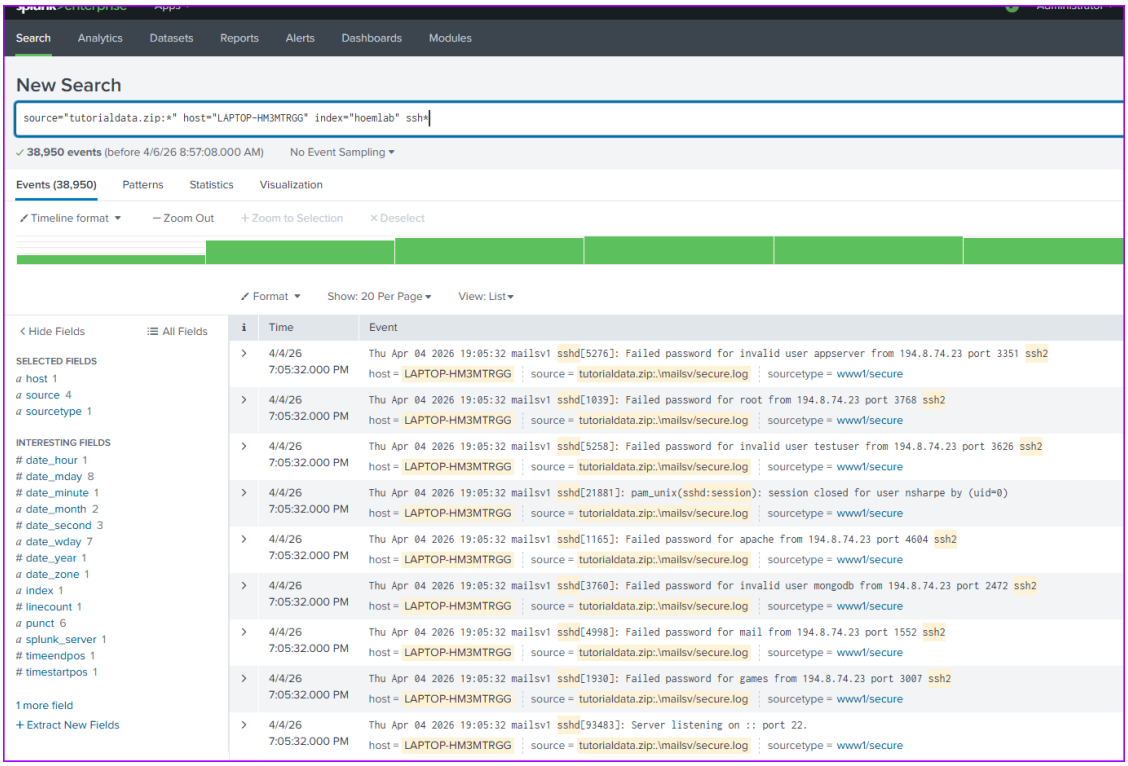

Figure 11 — 38,950 SSH events: failed logins mixed with legitimate session activity

Windows Event Viewer Investigation

Opened **Windows Event Viewer** on the host machine and navigated to **Windows Logs** → **Security**. Observed **28,911 security events**. Expanded Event ID **4799**:

Event ID	4799 — Security-enabled local group membership was enumerated
Subject	LAPTOP-HM3MTRGG\sugan (Logon ID: 0x131604)
Group	BUILTIN\Administrators (Group Domain: Builtin)
Process	C:\Windows\System32\taskhostw.exe (PID: 0x5f10)

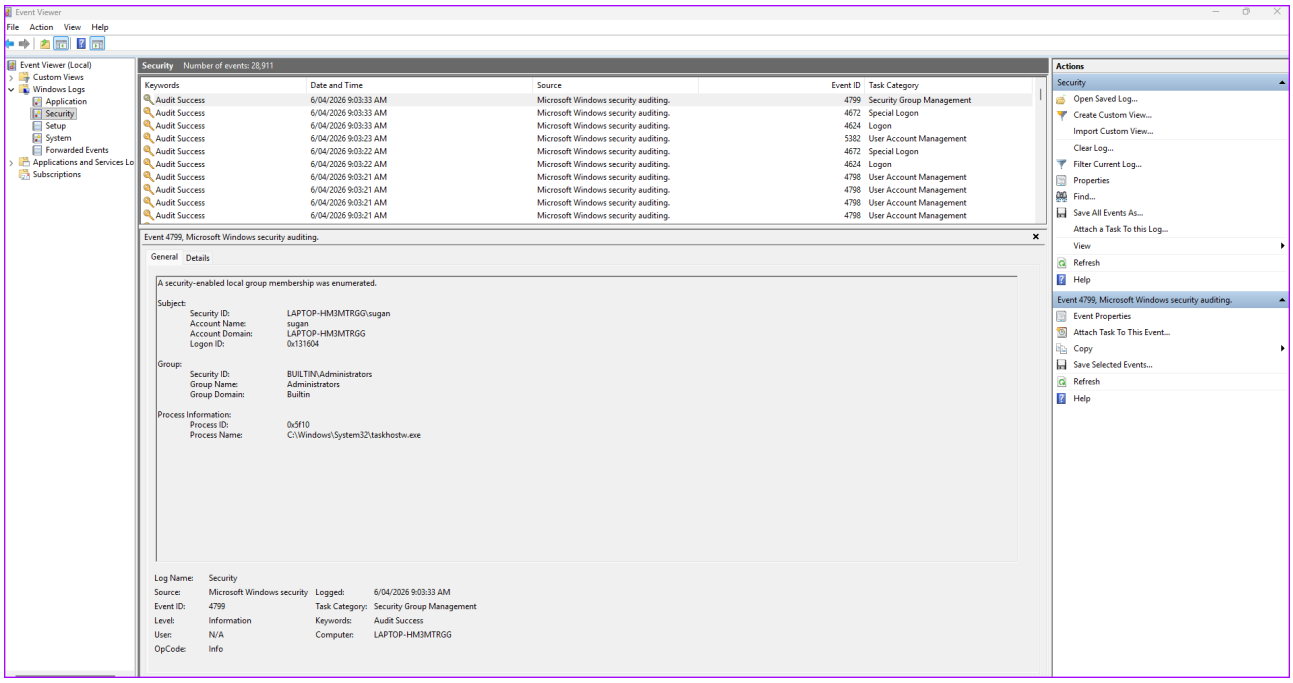

Figure 12 — Windows Event Viewer: 28,911 Security events with Event 4799 expanded (group enumeration by sugan)

Search 5 — All events in main index

```
index=main
```

Returned **59,056** → **107,257 events** as data continued loading. Sources included Perfmon:Network Interface, Perfmon:CPU Load, Perfmon:Available Memory, WinEventLog:Security, WinEventLog:Application, WinEventLog:System — confirming all Windows log sources were ingesting.

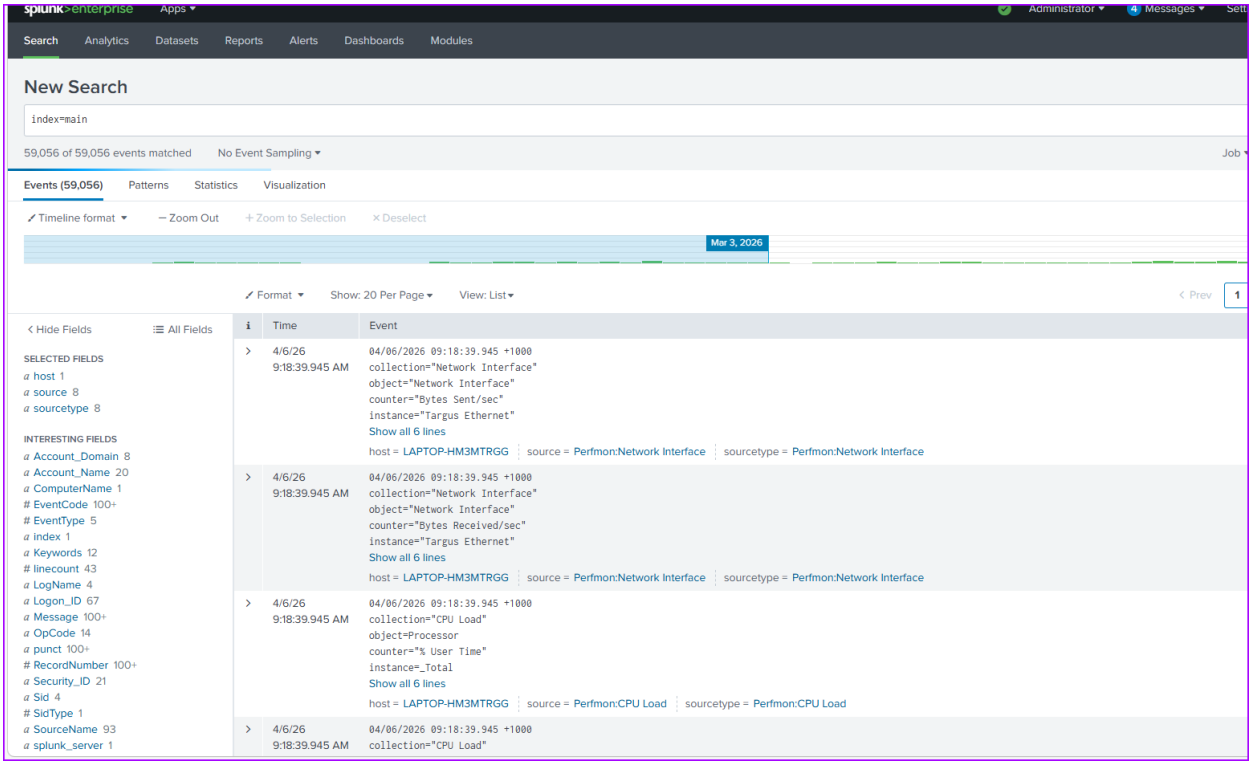

Figure 13 — index=main: 59,056 events showing Perfmon and WinEventLog sourcetypes in left panel

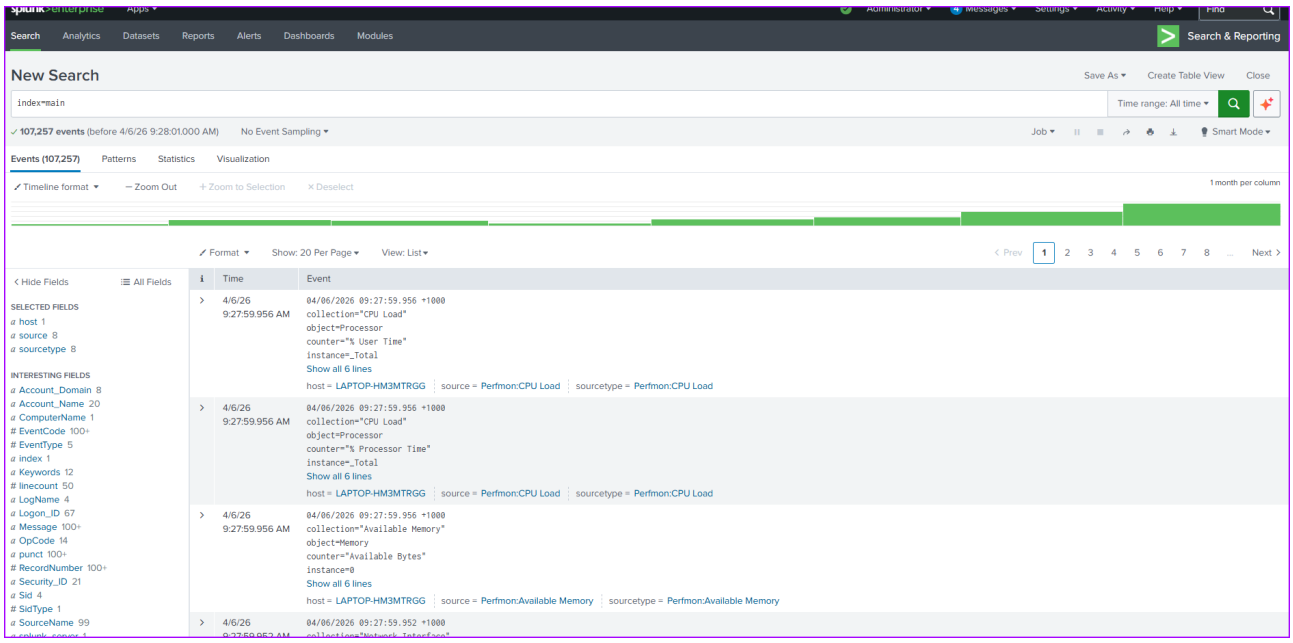

Figure 14 — index=main: 107,257 events with full sourcetype list visible after additional data loaded

Search 6 — Linux Authentication Log (auth.log)

```
source="auth.log" host="LAPTOP-HM3MTRGG" index="linux" sourcetype="Linux_auth"
```

Returned **5,548 events** from a Linux host (ip-10-77-20-248). Events included: sudo privilege escalation (ubuntu → root), SSH brute-force (Failed password for root from 73.231.4.205), authentication failures from sshd, and CRON session events.

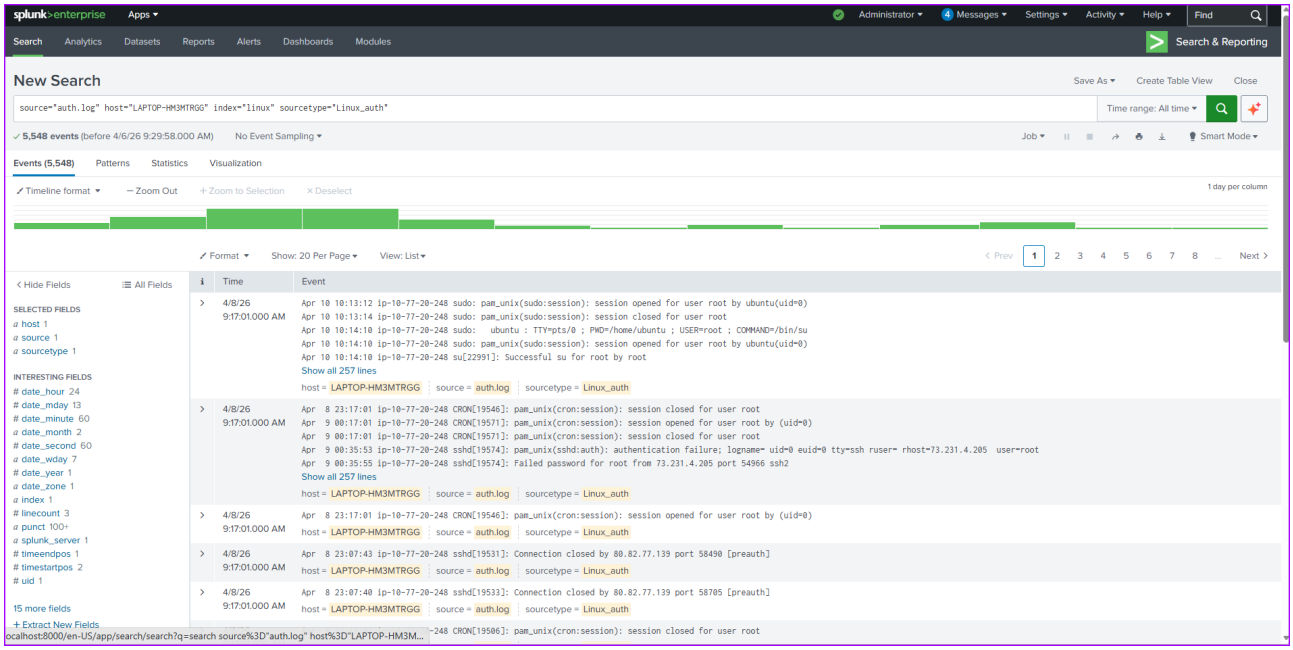

Figure 15 — 5,548 Linux auth events: sudo escalation, SSH failures from 73.231.4.205, CRON sessions

Search 7 — Live monitoring (index=main real-time)

```
index=main
```

Monitored live events as they streamed in — **72,158 events** visible during active ingestion. Confirmed Perfmon sourcetypes (CPU Load, Network Interface, Available Memory) updating in real time.

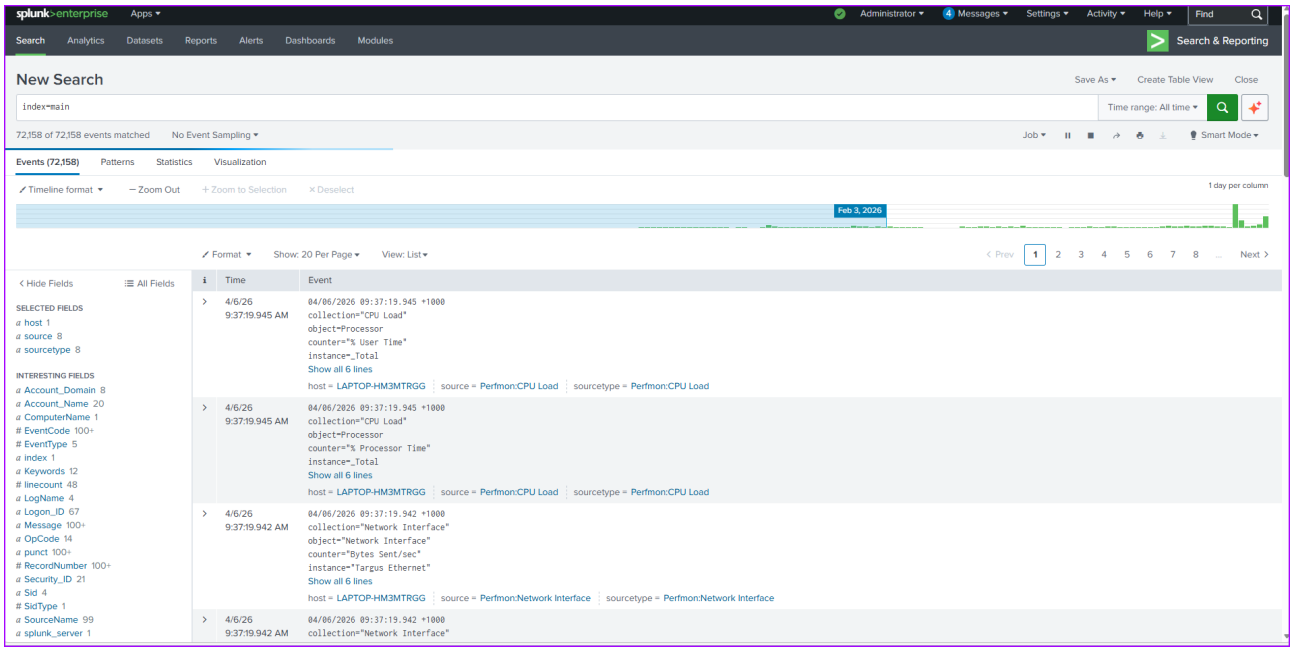

Figure 16 — Live index=main: 72,158 events actively ingesting with timeline showing data history

Search 8 — Exact phrase: 'Failed Password' (wrong case)

```
"Failed Password"
```

Returned **0 events**. Demonstrated that SPL exact phrase matching is case-sensitive. Capital 'P' in Password produced no results — a critical lesson for writing accurate detection queries.

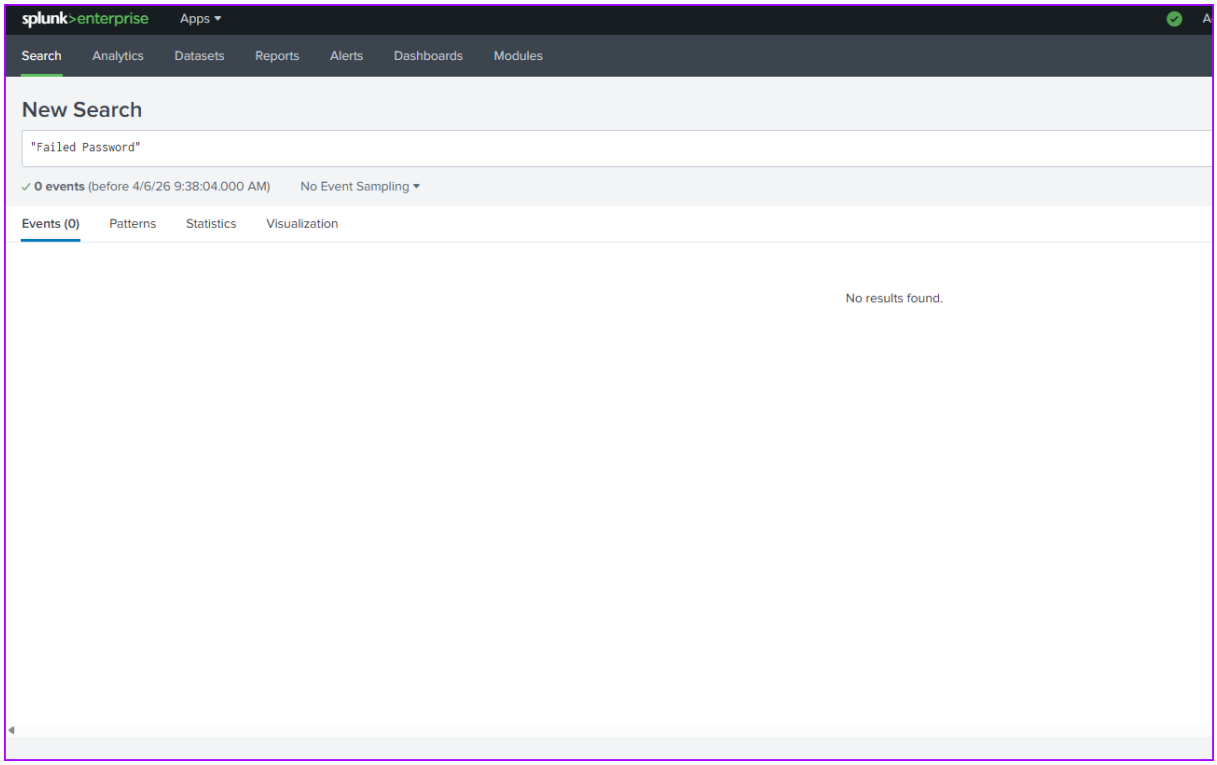

Figure 17 — "Failed Password" returns 0 events — SPL is case-sensitive for exact phrase matching

Search 9 — Exact phrase: 'Failed' (correct case)

```
"Failed"
```

Returned **13,628 events** across multiple sourcetypes: WinEventLog:Security (failed logon attempts), WinEventLog:Application (HP Display Control Service installation failed), WinEventLog:System (TPM hardware failure). Highlighted the importance of correct case in SPL queries.

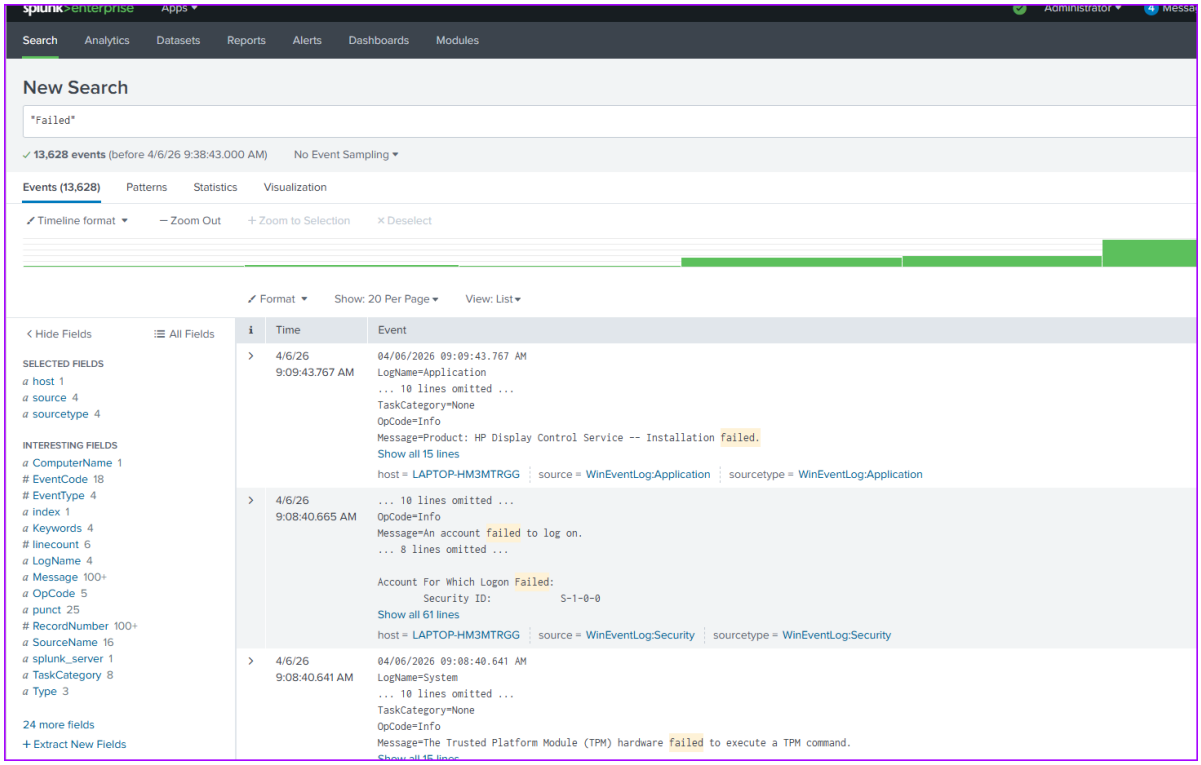

Figure 18 — "Failed" returns 13,628 events across Security, Application, and System logs

Search 10 — Linux Syslog: targeted failed password search

```
source="sample_auth_log.txt" host="LAPTOP-HM3MTRGG" index="sample_auth_log_1"
sourcetype="Linux Syslog" "Failed password"
```

Returned **6 events**. Multiple attackers targeting service accounts via SSH: mysql, oracle, postgres (from 185.234.217.22), guest (from 10.0.0.5), root and admin (from 192.168.1.10). Classic credential stuffing / automated scanning pattern across service account names.

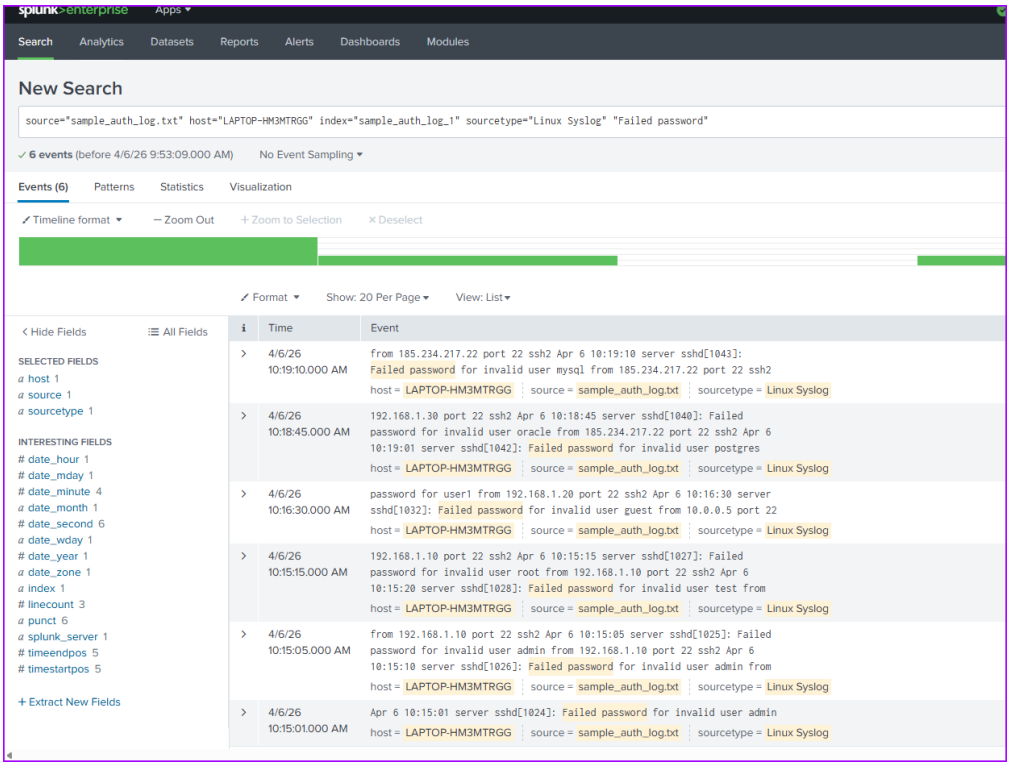

Figure 19 — 6 events in sample_auth_log_1: SSH failed passwords for mysql, oracle, postgres, guest, root, admin

■ Key Findings from Log Analysis

Finding	Detail
SSH Brute-force	IP 194.8.74.23 hammered mailsv1 sshd with 8+ invalid usernames (appserver, root, testuser, apache, mongodb, mail, games) — 33,253 matching events
Credential Stuffing	Multiple IPs (185.234.217.22, 192.168.1.10, 10.0.0.5) targeting service accounts: mysql, oracle, postgres, admin, root via SSH port 22
Privilege Escalation	Linux host ip-10-77-20-248: ubuntu user opened sudo session as root — potential post-compromise escalation
Group Enumeration	Windows Event 4799: user sugan enumerated BUILTIN\Administrators group membership via taskhostw.exe — reconnaissance indicator
SPL Case Sensitivity	"Failed Password" = 0 results vs "Failed" = 13,628 results — wrong case silently misses all matching events
Perfmon Monitoring	CPU Load, Network Interface, and Available Memory all ingesting successfully from LAPTOP-HM3MTRGG into main index

■ Results & Outcomes

Task	Result
Splunk Enterprise 10.2.2 installed, accessible at localhost:8000	■ Success
Receiving port 9997 configured and enabled	■ Success
tutorialdata.zip ingested into hoemlab index	■ Success
Windows Perfmon and Event Logs ingesting into main index	■ Success
Linux auth.log ingested into linux index (5,548 events)	■ Success
Linux Syslog (sample_auth_log.txt) into sample_auth_log_1	■ Success
SSH brute-force pattern identified in tutorialdata	■ Success
Credential stuffing pattern identified in sample_auth_log	■ Success
Windows Security Event 4799 investigated in Event Viewer	■ Success
SPL case sensitivity demonstrated and understood	■ Success

■ Key Takeaways

- **Data onboarding is step one** — Before any detection is possible, logs must be correctly indexed and searchable. Getting source, host, index, and sourcetype right is fundamental.
- **SPL is case-sensitive for exact phrases** — 'Failed Password' vs 'Failed' returned 0 vs 13,628 results. A detection query written with wrong case silently misses every event.
- **Brute-force attacks leave clear patterns** — One IP hammering 8+ usernames within seconds in the tutorialdata is exactly the kind of pattern SOC analysts hunt in SIEM tools.
- **Multiple log sources = better visibility** — Windows Event Logs, Perfmon metrics, and Linux auth logs combined give a far broader picture than any single source alone.
- **Targeted SPL searches are more powerful** — Adding index=, source=, sourcetype=, and host= dramatically narrows results and speeds up investigation.
- **Event IDs carry context** — Seeing Event 4799 (group membership enumeration) and knowing it signals potential reconnaissance separates a capable SOC analyst from someone just running searches.

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CompTIA Network+ • Log sources, syslog, network monitoring, SSH
CompTIA A+ • Windows Event Viewer, system logs, OS administration